ENGR 111
Introduction to Engineering
Class #13

Ethics and professional licensure

What the profession expects when the work affects other people

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

Engineering as a profession

Textbook: Landis Chapter 2.11. These are traits of learned professions.

Traits apply to more than engineering. Medicine, law, architecture, accounting: specialized judgment plus public trust.

Why Landis lists them. If the public cannot check the work, the profession needs standards, supervision, and discipline.

Junior engineers. You are not expected to carry this alone. You work under review while you build competence.

Codes of conduct. Some are society codes, some are employer rules, and some become state-board rules.

Recognition by law. Licensure is the legal version: who may offer, sign, or seal engineering work.

This is the bridge from student work to professional responsibility.

Morals, values, ethics

Textbook: Landis Chapter 8.6. Codes do not all come from the same place.

Morals

Personal right and wrong.

Shaped by culture, religion, family, and experience.

Not automatically a professional rule.

Example: refusing defense work.

Values

Priorities when good things compete.

Safety, cost, privacy, access, speed, sustainability.

Not automatically legally binding.

Example: a safer design that costs more.

Engineering ethics

Shared rules for professional behavior.

Sources: NSPE, IEEE, ASCE, ASME, state boards.

Binding when adopted by employer, society, or law.

Example: report a public safety risk.





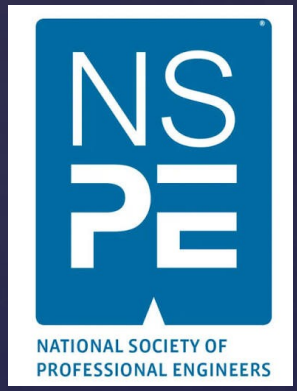


Question: when a code and your private morals disagree, what do you do?

NSPE ethics canons

NSPE is a professional society, not a licensing board. Oregon licenses PEs through OSBEELS.

Canon	Canon summary	Plain version
1	Safety, health, and welfare of the public	**Protect people first**
2	Work only in areas of competence	Know your limit
3	Objective and truthful public statements	Do not spin the facts
4	Faithful agent for employer or client	Represent who hired you
5	Avoid deceptive acts	No fraud, no false credentials
6	Honor, responsibility, lawfulness	Do not damage the profession



When the canons pull against each other, start with public safety.

Three duties, public first

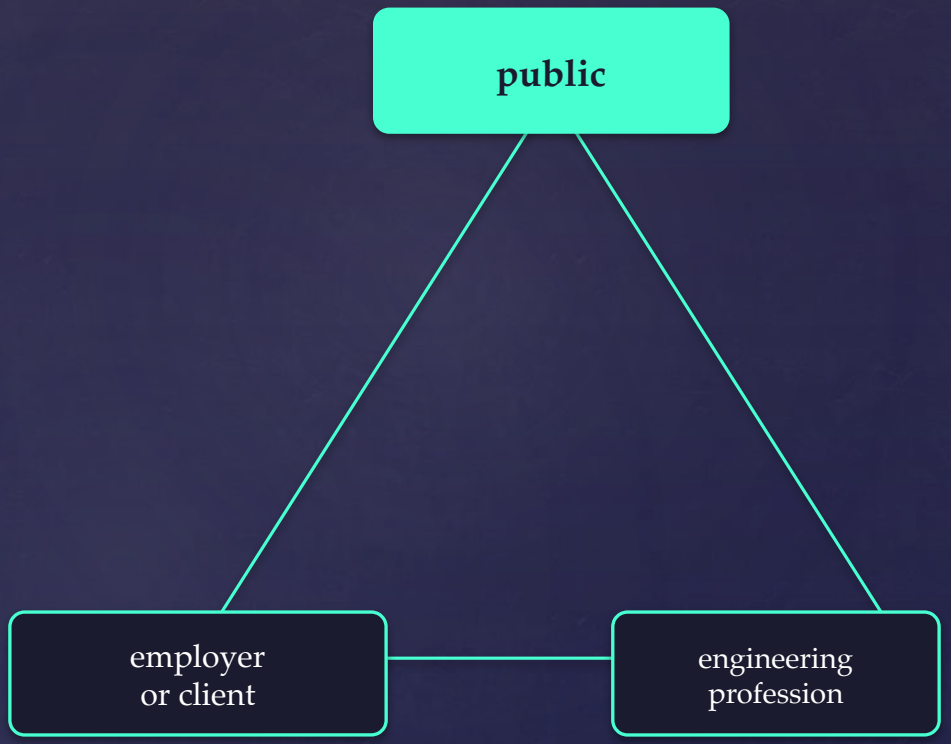
Engineers owe duties to more than one group. NSPE puts public safety above the others.

Public. People who may be hurt by the work, even if they never hired you.

Employer or client. The person or organization paying for the work; you owe honesty and confidentiality.

Engineering profession. The standards of the field: do not sign work you did not check.

Conflict example. A manager wants a safety concern softened before it goes to a customer.



public safety overrides the others

If life or property is endangered, report inside the organization and then to another authority if needed.

For real decisions, start with rules

A professional ethics problem is not just personal opinion.

Find the rules. Employer policy, project contract, law, state-board rules, and relevant professional codes.

Know your code. IEEE, NSPE, ASCE, ASME, and other societies write different codes, but safety and honesty recur.

Get the facts. Do not decide from rumor. Identify what is known, uncertain, and documented.

Use the chain. Ask a supervisor, mentor, ethics office, client contact, or regulator when the issue is serious.

Public safety. When people can be hurt, public protection overrides convenience, cost, and embarrassment.

In practice, start by reading the code that applies to your work.

Textbook (Landis): ethics method

1. Name the affected parties. Users, nearby public, client, employer, regulators, future maintainers.

2. List possible actions. Release, delay, redesign, warn, escalate, withdraw from the work.

3. Evaluate consequences. Short term and long term, including what happens if everyone does the same thing.

Selected textbook ethics checks

Ethics check	Question
Harm	Do the benefits outweigh the harms?
Reversibility	Would this still look fair if you traded places?
Colleague	What would a serious engineer say?
Publicity	How would this read in a public report?
Mirror	Would you be proud of the decision?

Landis is not a law or licensure rule. It is the textbook method for reasoning through a case.

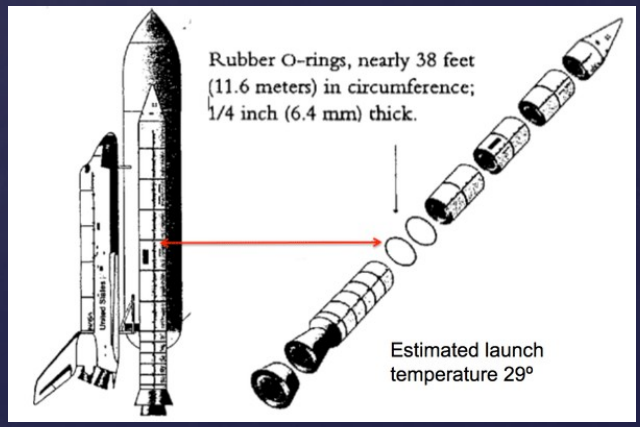
Case: Challenger launch (Jan 28, 1986)



NASA operated the Space Shuttle. Morton Thiokol was the contractor that built the solid rocket boosters. Challenger became a classic ethics case because earlier flights had shown O-ring erosion and blow-by, meaning hot gas leaked past rubber-like seals, but no booster had failed.



The night before launch, Thiokol engineers warned that the forecast cold was outside prior flight experience and recommended waiting until the O-rings were at least 53 F. NASA and Thiokol managers pressed for a launch decision; after an off-line management caucus, Thiokol reversed the no-launch recommendation.



Dilemma. Launch under schedule pressure with inconclusive data, or delay because safety had not been shown.

Warning. Cold could make seals slow to seat; prior erosion and gas leakage.

Authority. Contractor management, NASA project managers, launch readiness chain.

Public risk. Crew and everyone depending on the launch decision.

Suppose you are a Thiokol engineer in the launch-eve meeting. Your managers say NASA needs a decision tonight, the flight is already delayed, and your data are not conclusive. You think cold could stop the O-rings from sealing. What do you do?

Case: FIU bridge collapse (Mar 15, 2018)

Florida International University was building a pedestrian bridge over busy SW 8th Street in Miami. The main span had been moved into place, but the bridge was not open to pedestrians yet. Traffic was still passing underneath while construction work continued.



Before the collapse, cracks were visible near a critical connection. The project team discussed the cracks, but the road stayed open and work continued. NTSB later found design calculation errors, inadequate peer review, and failure to recognize the danger of the cracking.

Dilemma. Keep work and traffic moving, or close the road before the crack cause is certain.

Warning. Visible cracking near a critical connection; cause not resolved.

Authority. Engineer of record, contractor, FIU, and road/agency partners.

Public risk. Drivers, passengers, workers, and nearby public.

Suppose you are an engineer on the bridge team. Closing SW 8th Street will delay work, cost money, and embarrass the project. The cracks are real, but no one has proved collapse is imminent. Traffic is still below. What do you do?

When a PE license matters

PE licensure is not for every engineering job. It is mainly about public practice.

Public practice. Oregon generally requires registration to practice or offer professional engineering services to others.

Public-facing documents. Final drawings, specs, reports, maps, and plans for permits, clients, or agencies may need a PE seal.

Inside a company. Many EEs at places like Intel do not need a PE for product design under company responsibility.

Where it helps. Consulting, utilities, facilities, infrastructure, and building electrical work may require or prefer it.

Public safety. Safety still matters in product work. PE licensure is the legal accountability system for certain public-facing work.



Ask: am I offering engineering services, or submitting work someone else must rely on?

Signing, sealing, and consequences

A PE seal is a legal and professional certification, not a decorative stamp.

Active license. The signer is currently registered to practice in that jurisdiction.

Responsible charge. The PE had control over the engineering decisions, quality checks, and final work product.

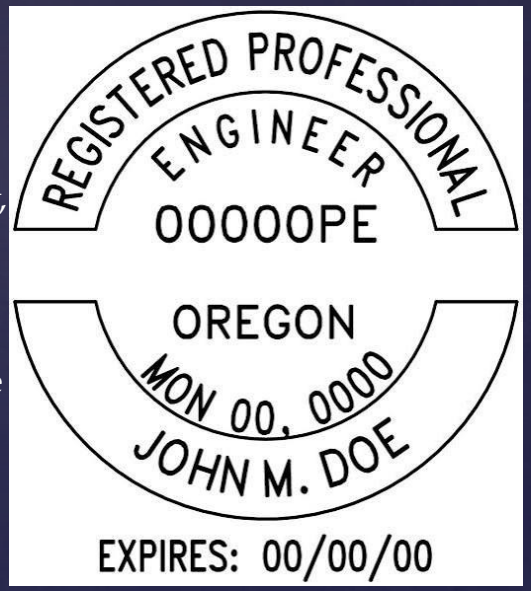
Supervision and review. Work may be done by a team, but the PE must direct, review, and accept it.

Standards. The PE accepts responsibility that the work meets professional practice standards.

Firm and individual. The firm may be sued, but the PE's license and public record are still exposed.

How trouble starts. Misusing a seal, working outside competence, deception, gross negligence, or failing to report discipline.

Consequences. Board complaint, public discipline, civil penalty, reprimand, suspension, revocation, job damage.



Signing says: I am the licensed professional accountable for this final engineering work.

Oregon PE pathway

Oregon State Board of Examiners for Engineering and Land Surveying (OSBEELS) reviews the license application.



EI = Engineering Intern title. After the FE you may register the EI title with Oregon. It is not MECOP, not a job, and not required for experience.

Experience is verified. Your PE application documents qualifying work; at least three registered professionals verify it and give you references/endorsement.

Exam order can vary. Oregon does not preapprove NCEES exams. Pass the required exams, then apply through MyOSBEELS.

Simple version: degree, FE, qualifying experience, PE exam, application, references.

Who controls PE licensure?

Licensure is state authority. Exams are national tools used by state boards.

Who	What they do	What they check
Oregon board (OSBEELS)	Grants Oregon PE licenses and can discipline licensees.	Application, experience, references, seal use, Oregon rules.
NCEES	Runs the FE and PE exams used by many states.	Exam registration, exam scores, records transfer.
References	Registered professionals verify your engineering experience.	Whether your work was real qualifying engineering practice.
Employer	May require a PE for a job, but does not issue the license.	Company performance, projects, and internal responsibility.

Simple version: Oregon grants the license; NCEES runs exams; references verify experience.